

1. Administrative & Study Identification

Full Study Title: PSMAcTION: A Phase II/III, Open-label, International, Multicenter, Randomized Study of AAA817 Versus Standard of Care in the Treatment of Adult Participants With PSMA Positive Metastatic Castration-resistant Prostate Cancer Who Progressed on or After [177Lu]Lu-PSMA Targeted Therapy **Short Title/Acronym:** PSMAcTION **Protocol Number:** CAAA817A12201 **NCT Number:** NCT06780670 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** AAA817 ([225Ac]Ac-PSMA-617) **Phase of Development:** Phase II/III **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations / Study Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** * *Phase II:* To determine the biochemical response rate, safety, and dose-limiting toxicity (DLT) to support the recommended Phase III dose of AAA817. * *Phase III:* To evaluate the efficacy—specifically radiographic progression-free survival (rPFS) and overall survival (OS)—and safety of the recommended dose of AAA817 compared to the investigator’s choice of standard of care (SoC). * **Secondary Objectives:** To evaluate progression-free survival (PFS), overall response rate (ORR), and overall safety/tolerability (including AEs and SAEs) across both phases.

2.2 Background & Rationale To address the unmet medical need in patients with PSMA-positive metastatic castration-resistant prostate cancer (mCRPC) who have already progressed on or after an androgen receptor pathway inhibitor (ARPI), taxane-based chemotherapy, and beta-emitting [177Lu]Lu-PSMA targeted therapy. AAA817 ([225Ac]Ac-PSMA-617) is a targeted alpha-emitting radio-ligand therapy. Actinium-225 has a higher linear energy transfer and shorter tissue path length than Lutetium-177, resulting in a higher ratio of double- to single-stranded DNA breaks, which may overcome resistance to prior therapies while potentially limiting off-target effects.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Investigator's choice of Standard of Care) * **Blinding:** Open-label * **Randomization:** Randomized

3.2 Planned Enrollment & Duration * **Target \$\$\$:** Approximately 432–443 adult participants (minimum of 12 for Phase II; ~420 for

Phase III) * **Number of Sites:** International, multicenter study (~54 global locations) * **Estimated Study Start:** January 17, 2025
* **Estimated Primary Completion:** June 27, 2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults ≥ 18 years of age with an ECOG performance status of 0 to 2. 2. Histopathological and/or cytological confirmation of adenocarcinoma of the prostate, with PSMA-positive disease assessed by PSMA PET/CT scan. 3. Prior treatments with an ARPI and taxane-based chemotherapy, and disease progression on or after [177Lu]Lu-PSMA targeted therapy. 4. Castrate level of serum/plasma testosterone (< 50 ng/dL or < 1.7 nmol/L) and ≥ 1 metastatic lesion present on baseline imaging (CT, MRI, or bone scan).

4.2 Exclusion Criteria 1. Use of any investigational agents within 28 days prior to randomization, or any prior use of 225Ac-based investigational compounds. 2. History of CNS metastases in patients who are neurologically unstable, symptomatic, or requiring corticosteroids; or concurrent acute kidney injury/chronic kidney disease. 3. Baseline xerostomia $\geq$ Grade 2; history of uncontrolled hypertension, myocardial infarction, angina, or CABG within 6 months prior to consent; or history of other known malignancies within the past 5 years.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** AAA817 ([225Ac]Ac-PSMA-617) given intravenously once every 8 weeks (1 cycle = 8 weeks). Phase II evaluates exploratory doses (Dose A and Dose B); Phase III utilizes the Recommended Phase 3 Dose. * **Route of Administration:** Intravenous (IV) infusion. * **Duration of Treatment:** Up to 6 cycles (approx. 48 weeks).

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard supportive care. For the control arm, treatments may include ARPIs, carboplatin, taxane-based chemotherapy, or radium-223. * **Prohibited:** Cross-over to 225Ac-PSMA-617 is strictly prohibited for patients randomized to the SoC arm. PARP inhibitors and immunotherapies (except sipuleucel-T) are excluded from the investigator's choice of standard of care in Phase III.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, PSMA-PET/CT scan, Conventional Imaging, Medical History, Labs
Baseline	Day 0	Randomization, First Dose
Treatment Cycles	Every 8 weeks	IV Infusion of AAA817 or SoC, Safety Labs, PSA

levels, Efficacy/Imaging Assessments | | **End of Treatment** | 30 to 56 days post-last dose | Safety Follow-up: 56 days post-last dose for AAA817 arm, 30 days (+7) for SoC arm | | **Long-Term Follow-up** | Up to 5 years | OS monitoring, ongoing safety, and subsequent anti-neoplastic therapy tracking |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** * *Phase II:* Biochemical response rate, dose-limiting toxicities (DLT), and safety. * *Phase III:* Radiographic progression-free survival (rPFS) and Overall Survival (OS). * **Measurement Method:** Conventional imaging (CT/MRI/bone scans) evaluated by central/investigator review, and serologic PSA level evaluations.

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) * Overall Response Rate (ORR) * Incidence of Adverse Events (AEs), Serious Adverse Events (SAEs), and safety profiles including targeted tracking of xerostomia.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous monitoring of AEs/SAEs from randomization through the safety follow-up window, with a specific focus on radioligand toxicities (e.g., salivary gland toxicity/xerostomia, hematologic parameters, and renal function).
 - **Serious Adverse Events (SAEs):** Reported promptly per standard protocol; safety follow-up extends up to 56 days post-treatment.
 - **Data Monitoring Committee (DMC):** Utilized for interim safety evaluations, particularly to select the recommended Phase III dose based on Phase II dose-limiting toxicity (DLT) data.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy endpoints (rPFS and OS). Safety Set for all patients receiving at least one dose of the study drug.
 - **Sample Size Justification:** Approximately 432-443 total patients. Phase III (\$N \approx 420\$) is powered to detect statistically significant improvements in dual primary endpoints (rPFS and OS) between the AAA817 and SoC arms.
 - **Handling of Missing Data:** Standard oncology imputation; right-censoring utilized for time-to-event survival metrics (rPFS, OS, PFS).
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Rigorous onsite and remote clinical monitoring overseen by Novartis Pharmaceuticals, coupled with central independent reviews of imaging data.
 - **Audit Trail:** Validated eCRF locking, tracking of protocol deviations, and standardized data clarification mechanisms.
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11. Study Identifiers & Governance

- **Primary ID:** CAAA817A12201
 - **Registry Number:** NCT06780670
 - **Sponsor/Lead Organization:** Novartis Pharmaceuticals
 - **Study Title:** PSMACTION
 - **Official Regulatory Title:** A Phase II/III, Open-label, International, Multicenter, Randomized Study of AAA817 Versus Standard of Care in the Treatment of Adult Participants With PSMA Positive Metastatic Castration-resistant Prostate Cancer Who Progressed on or After [177Lu]Lu-PSMA Targeted Therapy
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12. Clinical Framework (mCRPC Specific)

- **Primary Condition:** Metastatic Castration-resistant Prostate Cancer (mCRPC)
 - **Diagnosis Threshold:** PSMA-positive disease assessed by PET/CT; castrate level of serum/plasma testosterone (< 50 ng/dL).
 - **Medical Methodology:** Targeted Alpha-emitting Radio-ligand Therapy vs. Investigator's Choice of Standard of Care.
 - **Optimization Target:** Radiographic Progression-Free Survival (rPFS) and Overall Survival (OS) extensions.
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13. Study Procedures & Methodology

- **Management Pathway:** Intravenous targeted alpha-radioligand therapy infusion mapping.
 - **Education Protocol (HCP):** Protocol training on precise handling, administration, and radiation safety protocols for Actinium-225 based therapies.
 - **Education Protocol (Patient):** Education on radiation safety following isotopic infusion, tracking of dry mouth (xerostomia), and necessary fluid intake.
 - **Quality Audit Mechanism:** Independent central review of PSMA PET/CT scans and conventional imaging to verify progression criteria.
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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to 48 weeks of active treatment; up to 5 years of long-term survival follow-up.
- **Onsite Visit Cadence:** Every 8 weeks coinciding with designated treatment cycles.
- **Remote Monitoring Frequency:** Between treatment cycles and during the long-term OS tracking period.
- **Checklist Review Interval:** Rigorous safety check at 30 days (+7) or 56 days post-treatment cessation.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					